Classifying EEG-Based Motor Imagery for Drone Control Using Artificial Neural Network

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Abstract— **The progress in brain-computer interface (BCI) technology has created new possibilities for controlling external devices through electroencephalogram (EEG) signals. This study investigates the classification of EEG-based motor imagery (MI) signals for drone control using artificial neural networks (ANN). EEG data from six participants was collected using the Emotiv Epoc X device, focusing on MI related to drone movements. Feature extraction was performed using the discrete wavelet transform (DWT), followed by classification with a two-layer ANN with statistical parameters as input. The results indicate the significance of the ANN regression model in classifying motor imagery, although variations in model performance were observed across subjects. While overfitting was noted in one subject, indicating the need for validation sets, the models for other subjects showed good generalization. The ANN model achieved high accuracy in classifying movements "take-off" (98%) and "forward" (99%), but struggled with the "turn-left" (82%), suggesting the need for further model adjustments or additional data to improve generalization.**

Keywords—BCI, EEG-Motor Imagery, ANN, wavelet transform, drone

I. Introduction

Brain-computer interface (BCI) techniques base a direct connection between the brain and a computer, making a communication channel that bypasses the peripheral nervous approach. BCI utilizes neurofeedback technology to capture and interpret electroencephalogram (EEG) signals, allowing individuals to control external devices based on their preferences. Many studies have examined brain-computer interfaces to detect, identify, and apply to various areas such as communication, education, entertainment, and medical applications [1-3]. The most generally used EEG responses In BCI designs are Steady State Visual Evoked Potentials (SSVEP), Sensorimotor Rhythms (SMRs), Event Related Potentials (ERPs), Event Related Desynchronization / Synchronization (ERD/ERS), and Motor Imagery (MI) [4].

BCI based on MI means that there is no actual physical behavior involved, but rather thoughts in the brain are used to imagine bodily movements. MI activates brain areas that are similar to those activated during actual physical movement. This technique has comprehensive prospects for applicability. So far, BCI based on MI EEG signals has numerous applications in various fields, such as brain-controlled dual-arm robots, controlling drones, driving virtual cars, and fatigue monitoring. Decoding imagined movements could be invaluable in these fields [5-6].

Unmanned aerial vehicles (UAVs), or drones, have gained attention for their role in aerial operations, especially when piloted flights are not feasible. Recently, drones controlled by BCI technology have become one of the numerous intriguing topics. Several Respon Wang et al. [8] developed a head-mounted BCI system using SSVEP for three-dimensional quadcopter navigation with visual feedback. LaFleur et al. [9] showed quadcopter control in three-dimensional space employing noninvasive scalp EEG signals established on motor imagery EEG response [10].

In this paper, the experiment used MI-based BCI, in which pattern recognition systems affect three essential processes: preprocessing of the EEG signal, feature extraction, and classification, with feature extraction being pivotal [7]. Signal processing is employed for analyzing EEG data in several steps. i.e., feature extraction using wavelet transform (WT) to analyze signals in time fields, decomposition, and compression signals in which non-linear EEG signal characteristics need to be analyzed using function wavelets. Additionally, the method of classifying using neural network regression has demonstrated its capacity to extract spatial and temporal features from MI data. This approach can effectively retrieve valuable features from both shallow and deep models, indicating that significant features can be obtained at different levels [10]. The scheme of the experiment shown in Fig. 1.

For the MI classification method, several classification algorithms were presented based on the study. The most employed method is Emotiv's software built-in classification model, which three studies have used [11-13], and the LDA algorithm that is used in two papers [14-15]. Reference [16] utilized a logistic regression model to classify the three mental commands recorded by the participants. Reference [17] developed an ANN to group the four available controls of their system. Especially for ANN, it requires careful parameter selection, which varies based on the task and subjects. Optimizing EEG input data and channel selection is crucial for efficient ANN-based BCIs. Training can improve EEG features and make recognition easier for ANN-based algorithms. This is because many BCI studies involve trained subjects. After all, it's harder to classify brain activity patterns in untrained subjects during motor imagery [18-19]. The ultimate objective of this study is to develop a prototype drone

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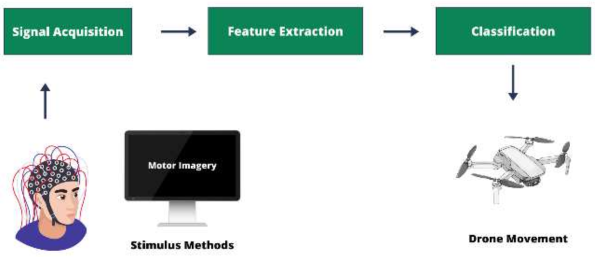

Fig. 1. The experimental design.

for defense applications utilizing EEG-based motor imagery (EEG-MI) for drone control.

II. METHOD

A. Data Acquisition

Six subjects were selected for this experiment, consisting of four males and two females. EEG signals were recorded by Emotiv Epoc x with Sampling Rate 256 Hz and 14 channels: AF3, F7, F3, FC5, T7, P7, O1, O2, P8, T8, FC6, F4, F8, and AF4. Before the experiment, participants were asked to complete a questionnaire about their current health status. Following this, they were seated in a chair, where they listened to instructions from the staff and adhered to rules, including minimizing unnecessary body movements. The experimental setup is shown in Fig. 2, in which the subject will be given a motor imagery stimulus. The MI design employed in this experiment utilizes videos depicting drone movements such as take-off, landing, turning right, turning left, and forward, with each video having a duration of 5 seconds and loop three times. The design of MI is shown in Fig. 3.

B. Feature Extraction

Wavelet transform theory has diverse applications in digital signal processing and plays a crucial role in the analysis of time-domain signals by providing information about both frequency and time within the signals [20]. Non-

(a) (b)

Fig. 2. Experimental setup: (a) EEG signal recording, (b) electrode in Emotiv Epoc X.

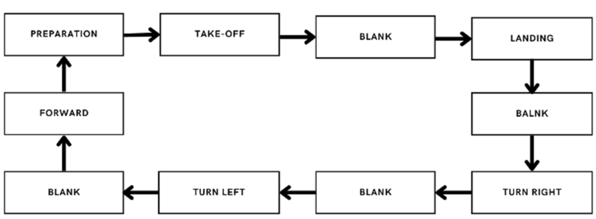

Fig. 3. The design of motor imagery stimulus.

linear EEG signal characteristics need to be analyzed using function wavelets ($\psi_{a,b}$,*mother wavelet),*

$$\psi_{a,b} = \frac{1}{\sqrt{a}}\psi\left(\frac{t-b}{a}\right) \qquad (1)$$

where a, b ∈ R, a > 0, and R is the *wavelet space.* The *mother wavelet* function serves as the foundation for generating all the basic wavelet functions, which are designed to match the specific characteristics of the target function. This process involves utilizing the mother wavelet to derive a complete set of wavelet functions that collectively capture the desired features and patterns of the function being analyzed.

The Discrete Wavelet Transform (DWT) is a mathematical technique used for analyzing signals and images. DWT has two sets of functions called scaling functions and wavelet functions associated with low pass filters and high pass filters. The DWT process can be seen in Fig. 4 [21,22].

C. Classification

An artificial neural network (ANN) is an interconnected group of nodes used in machine learning. ANNs are based on the neural structure of the brain and are made up of interconnected processing elements (neurons). They are powerful tools for classification, pattern recognition, prediction, and pattern completion, inspired by biological neurons. The classification method used ANN with a two-layer feed-forward structure as shown in Fig. 5 where This structure illustrates how inputs are processed through weighted connections and activation functions in the hidden layer and then transformed into outputs in the output layer. The network learns by adjusting the weights and biases during training to minimize errors in output predictions. The ANN structure featured two transfer functions, with the first output neuron in the hidden layer displayed in (2)

$$a^1 = f^1(IW_p + b^1) \qquad (2)$$

In this context, a^1 denotes the output vector from the input layer, while p is an input vector of length n. The input weight matrix is represented by LW, f^1 is the transfer function applied in the hidden layer, and b^l is the bias vector for the hidden layer.

The first output neuron in the output layer, as shown in Fig. 5, corresponds to (3). Here, a^2 represents the output vector from the output layer; LW denotes the weight matrix

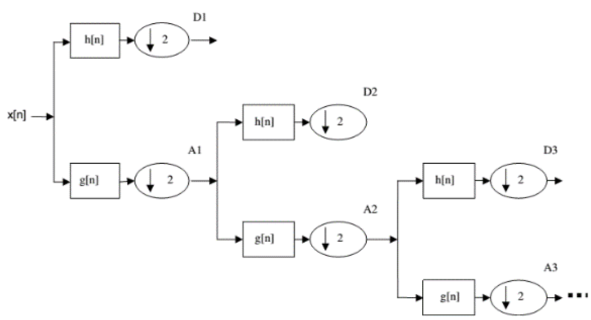

Fig. 4. Process decomposition of DWT.

associated with this layer; $f^2$is the layer's transfer function; and b^2 is the bias vector for the output layer [23].

$$a^2 = f^2(LW\ (f^1(IW_p + b^1)) + b^2) \tag{3}$$

III. Experimental Result

Fig. 6 presents the raw EEG signals from the first subject. The T7 and T8 channels have been excluded due to the minimal signal obtained from them, which is attributed to interference with the Emotiv device. The scheme of EEG signal processing is shown in Fig. 7. The data will be processed using detrending, a smooth function, for example a low order polynomial, is fit to the data and then subtracted from it. Detrending removes long-term trends and systematic patterns from data, making it easier to analyze smaller fluctuations and noise. It improves the visibility of relevant patterns and variations by focusing on changes rather than long-term shifts.

In the next step, as shown in Fig. 7, the data will be extracted using wavelet analysis. The choice of wavelet and the number of decomposition levels in the Discrete Wavelet Transform (DWT) will be tailored to fit the desired signal characteristics. For this research, the 'coif5' wavelet type is used for EEG signal processing, as illustrated in Fig. 8. The DWT features are used to obtain statistical parameters that serve as inputs for ANN classification. These statistical parameters, i.e mean, standard deviation, and average, play a crucial role in capturing the inherent patterns within the EEG signals. By extracting these features, the model can better distinguish between different motor imagery tasks. The selection of appropriate statistical parameters is critical, as it directly influences the model's ability to generalize across different subjects and motor imagery tasks. Incorporating a comprehensive set of statistical features not only enhances the

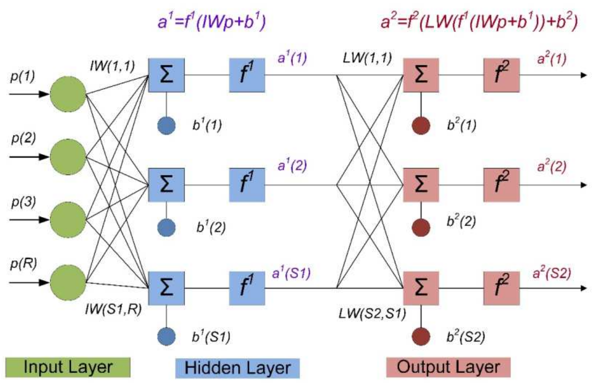

Fig. 5. Two layer feedfordward network.

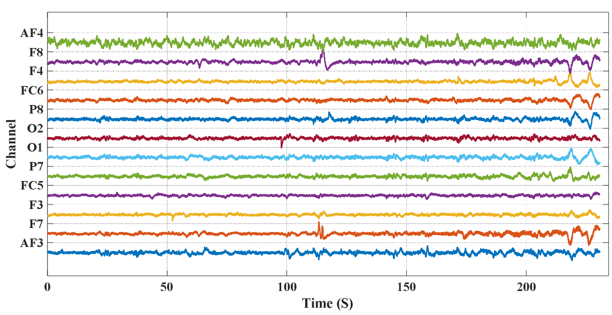

Fig. 6. Raw data EEG signals.

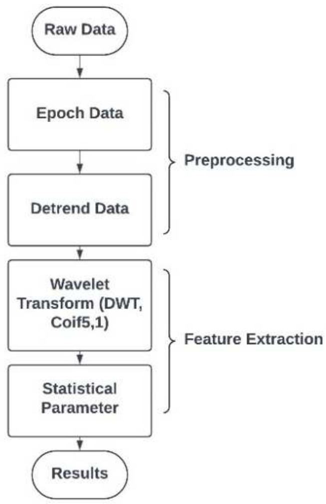

Fig. 7. The scheme of EEG signal processing.

ANN's performance but also aids in mitigating issues like overfitting by providing a more robust representation of the underlying signal characteristics. This careful extraction and selection process ensures that the input features are both meaningful and representative of the complex EEG data. In Fig. 9, the characteristic signal for each drone movement is: landing, forward, turning right, turning left, and landing.

Based on Fig. 10. The neural network trained on EEG Motor Imagery data shows good learning progress. This is clear from the quick drop in training errors and the steady performance on the validation and test sets. The fact that the validation and test errors remain stable over time suggests that the model isn't just memorizing the training data but is actually learning to generalize well to new data. This makes it a dependable model for classifying EEG Motor Imagery signals. Fig. 11. This graph shows which features of the EEG

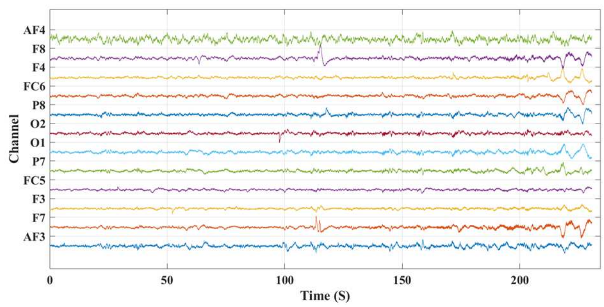

Fig. 8. The result of DWT.

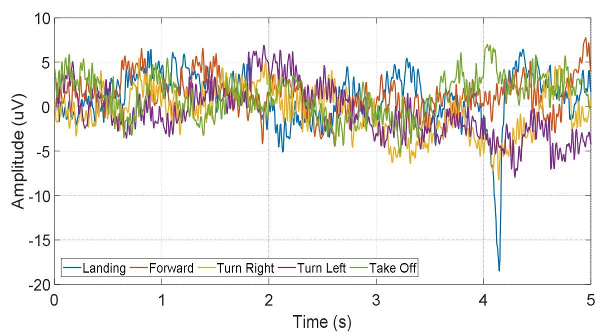

Fig. 9. The characteristic signal for each drone movement.

data the neural network relied on the most for making its predictions. One feature clearly stands out as the most important across all stages—training, validation, and testing. This suggests that the model is focusing on the key information, which likely helps it perform well on new, unseen data.

Fig. 12. demonstrates how well the Artificial Neural Network regression model performs in classifying EEG motor imagery data across all subjects. It includes four scatter plots, each representing a different dataset: training, validation, testing, and a combined dataset of all subjects.

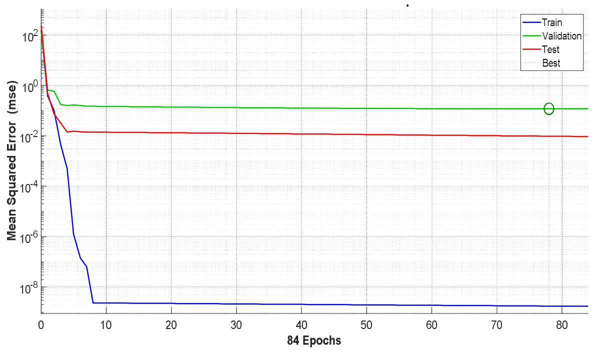

Fig. 10. Performance data of ANN classification results from all subjects.

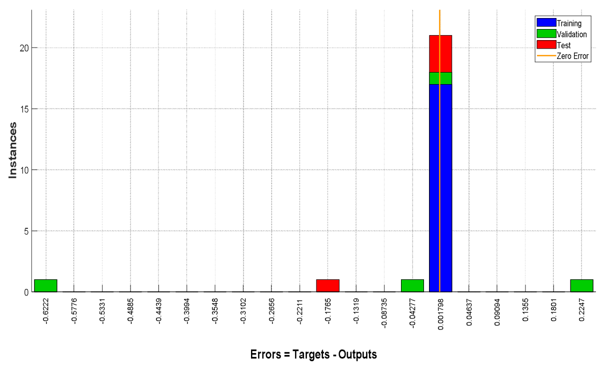

Fig. 11. Training State of ANN classification results from all subjects.

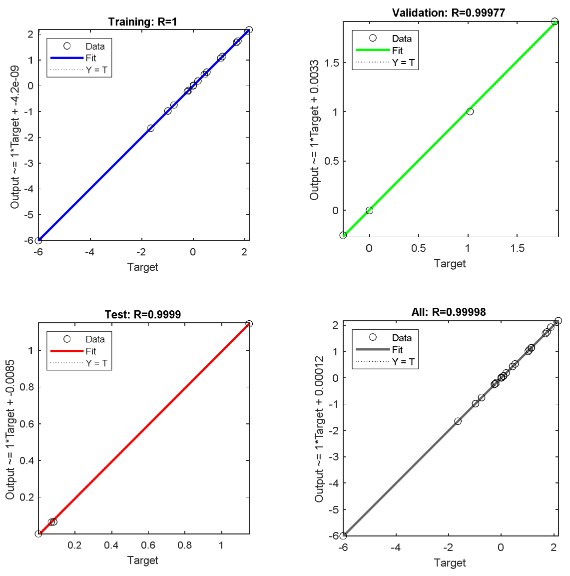

Fig. 12. Regresion of ANN classification results from all subjects.

In Table I. The study shows that ANN models perform differently across subjects, which is expected in EEG-based brain-computer interface research because everyone's brain patterns are unique. For example, Subject S2 experienced overfitting, where the model worked great on the training data but didn't do well on new data. This highlights why it's crucial to use validation sets to keep an eye on the model's performance and avoid overfitting. On the other hand, Subjects S1, S3, S4, and S6 had models that generalized well, meaning the features they learned worked well on new data too. Testing accuracy is really important because it gives the best idea of how the model will perform on data it hasn't seen before. If there's a big drop in testing accuracy compared to training and validation accuracy, it’s a warning sign that the model might not be ready for real-world use. The differences in model performance could be due to individual brain patterns, the quality of the EEG data, the features chosen for training, and how the hyperparameters were tuned, all of which play a role in how well the model generalizes.

The Table II. summarizes the accuracy of an ANN model used to classify different drone movements based on EEG Motor Imagery data. The performance metrics include accuracy during the Training, Validation, and Testing phases, along with an overall accuracy for each movement. The ANN model performs well in most cases, especially for "Take-Off" and "Forward" movements, which have very high overall accuracy. However, the "Turn-Left" movement is more challenging, as shown by a significant drop in Validation accuracy. This indicates that while the model can classify most movements effectively, it struggles with some, likely due to overfitting or less clear EEG data features. Improving the model's generalization, particularly for "Turn-Left," might require further adjustments or additional data

IV. Conclusions

The study on classification methods using artificial neural networks (ANN) was conducted and presented. EEG-based motor imagery signals corresponding to five drone directions

TABLE I. ANN Classification Results

No.	Subjek	Accuracy (%)			
		Training	*Validation*	*Testing*	*Overall*
1	S1	100	98	99	**99**
2	S2	99	61	78	**79**
3	S3	100	100	99	**100**
4	S4	100	100	99	**100**
5	S5	99	99	71	**90**
6	S6	99	99	99	**99**

TABLE II. ANN Classification Results In Each Drone Movement.

Drone Movement	Accuracy (%)			
	Training	*Validation*	*Testing*	*Overall*
Landing	100	98	81	**93**
Take-Off	100	99	94	**98**
Turn-Right	86	99	87	**91**
Turn-Left	100	57	90	**82**
Foward	100	97	99	**99**

from six subjects were processed and analyzed using wavelet transform. This technique extracts statistical parameters for input classification. The overall classification results across all subjects demonstrated the effectiveness of the artificial neural network regression model in classifying EEG motor imagery. However, the study also highlighted variations in model performance among different subjects in EEG-based brain-computer interface research. Overfitting was observed in Subject S2 (72%), indicating the need for validation sets to mitigate this issue. In contrast, the models for Subjects S1 (99%) , S3 (100%), S4 (100%), and S6 (99%) showed good generalization. The accuracy of the ANN model in classifying various drone movements based on EEG motor imagery data was generally high. The model performed particularly well for "take-off" (98%) and "forward" (99%), achieving high accuracy. However, the "turn-left" (82%) movement proved more challenging, with a noticeable drop in validation accuracy. This suggests that improving the model's generalization, especially for the "turn-left" movement, may require further adjustments or additional data.

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